Scientific event: Biodiversity Dynamics in Coevolutionary Meta-Ecosystems

Contributions of the Invitees

Dr. Cecilia S. Andreazzi

Dr. Andreazzi will contribute to the scientific event with her *mathematical modeling expertise* in complex networks of interactions. Dr. Andreazzi will be co-leading the modeling team together with Dr. Astegiano to build a mechanistic "*Coevolutionary meta-ecosystem model*" containing many interacting species in spatially explicit landscapes. The extensions to be made to the mechanistic modeling during the scientific event will be driven by the empirical data contributed by other members of the team (see below the contributed data from Dr. Astegiano, Dr. Becks, Dr. Heleno, and Dr. Sabatino). The main extensions to the "*Coevolutionary meta-ecosystem model*" to be discussed during the scientific event will be

1. adding distribution of traits for each species following the empirical distributions
2. extending the model to the sampled empirical spatial network, and
3. running a prototype model under different scenarios to analyze their stability properties.

Dr. Matias Arim

Dr. Arim will contribute with his expertise in *network metrics* to quantify the effect of spatial arrangements and migration scenarios to disentangle the interplay between coevolutionary dynamics at the local and the landscape level. Dr. Arim will work closely with the group in charge of implementing the prototyping model to explore the role of spatial and metacommunity network structure on coevolutionary dynamics. The main tasks will be

1. informing about the spatial network metrics that have been reported to influence biodiversity dynamics, and
2. advising about how the existing spatial network algorithms could be merged to the main coevolutionary meta-ecosystem model.

Dr. Julia Astegiano

Dr. Astegiano will contribute both with her expertise in *modeling network patterns* in spatially explicit ecological networks and with her own dataset. Dr. Astegiano will work together with Dr. Andreazzi, Dr. Huang and Dr. Massol in building the prototype of the "*Coevolutionary meta-ecosystem model*". Specifically, she will contribute to implementing the functional traits in the coevolutionary model to explore the functional properties at species and at the meta-ecosystem level. Dr. Astegiano's will also contribute with her own dataset containing plant-pollinator interactions from nine different agricultural landscapes in Argentina. Currently there are approx. 1.5k interactions across these nine sites following 800m of plant-pollinator transects for each site.

Dr. Lutz Becks

Dr. Becks will contribute with his expertise in *statistical analysis of micro- and meso-cosm host-virus coevolution experiments*. Dr. Becks will also contribute with his own temporal and spatial data containing algae-virus infectivity-resistance trait coevolution. Dr. Becks' data will complement the data contributed by field community ecologists enriching the "Coevolutionary empirical patterns" database (see below Dr. Heleno's contribution). This will allow us to analyze coevolutionary patterns in both types of data (i.e. field and experimental data) to explore their similarities and differences.

Dr. Ruben Heleno

Dr. Heleno will contribute both with his expertise in *analyzing network patterns* in ecological networks and with his own dataset on *Galapagos* collected approx. during the last ten years. Dr. Heleno's dataset contains plant-pollinator-seed dispersers for about 5k interactions across many of the islands of the Galapagos Archipielago. Dr. Heleno will be in charge of integrating and analyzing the datasets during the scientific event, the "Coevolutionary empirical patterns" database. Specifically, Dr. Heleno will lead exploratory analysis to explore the similarities among trait distributions across the datasets, the isolation by distance network patterns, and sampling bias associated with each dataset. He will inform the modeling team about how we could parametrize the model prototype using empirically-driven scenarios.

Dr. Francois Massol

Dr. Massol will contribute with his expertise in *modeling evolutionary and ecological processes in spatially explicit landscapes*. He will team up with Dr. de Andreazzi, Dr. Astegiano and Dr. Huang in setting up the "*Coevolutionary meta-ecosystem model."* He will take the lead during the workshop to help parametrize the prototyping model using the empirical and spatial trait-based data. He will advise about methods to analyze the stability properties of the "*Coevolutionary meta-ecosystem model"* and how the different scenarios to explore could influence co-extinctions in meta-ecosystems.

Dr. Malena Sabatino

Dr. Sabatino will contribute both with her expertise in statistical analysis of *network patterns* in spatially explicit ecological networks and with her own dataset. Dr. Sabatino dataset contains plant-pollinator interactions for about 6k individual interactions with a total of 96 plant and 172 pollinator species across many "sierras." Dr. Sabatino will team up with Dr. Heleno for the exploratory analysis of the "Coevolutionary empirical patterns" database.

Dr. Weini Huang

Dr. Huang will contribute with her expertise in modeling host-parasite coevolution. Dr. Huang will team up with Dr. Andreazzi, Dr. Astegiano, and Dr. Massol in building the empirically-driven prototype of the "*Coevolutionary meta-ecosystem model*". Specifically, she will contribute to implementing the deterministic version of the model whose outputs will be used to test the processes governing the large population sizes of the host-virus coevolution experiments.